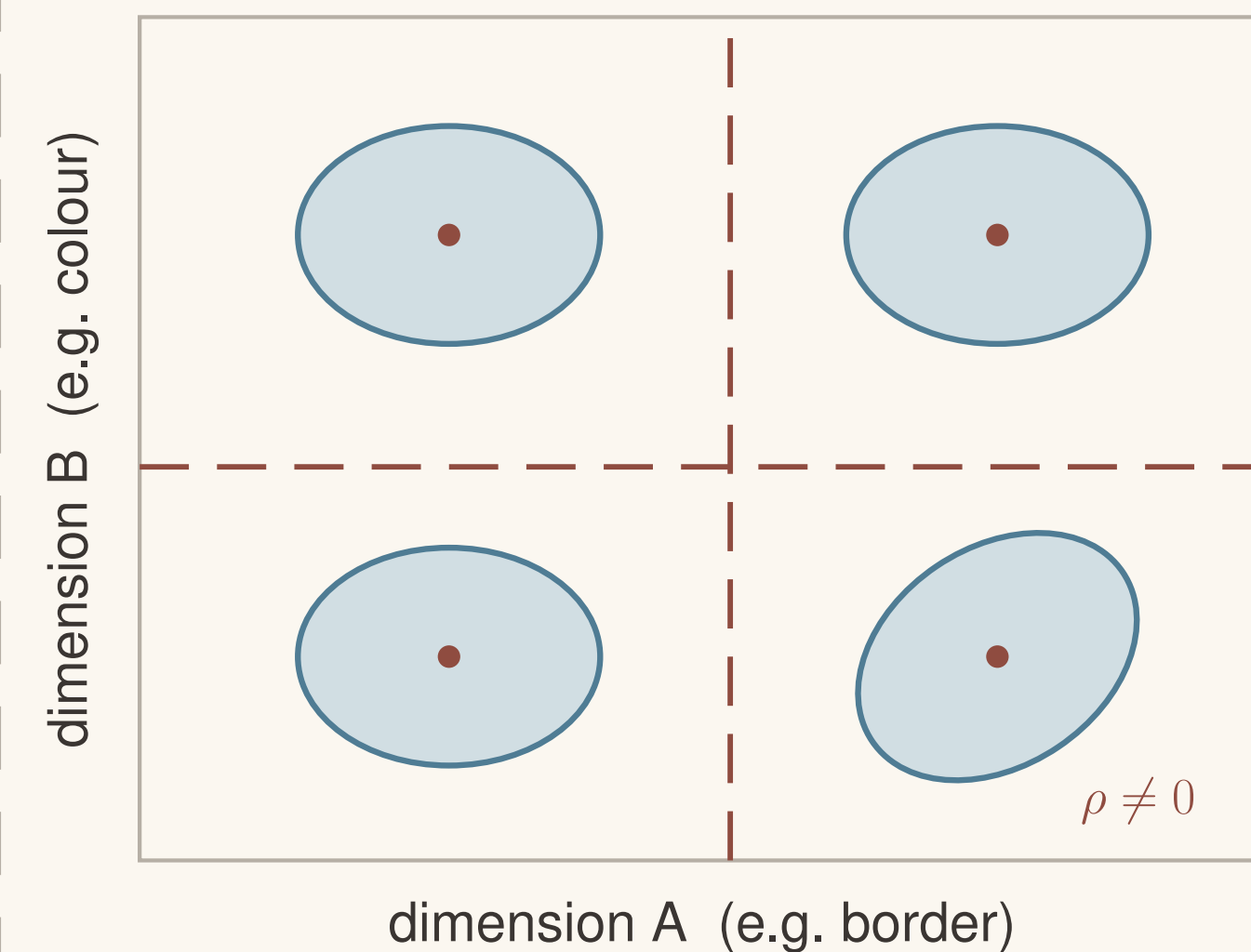


OFFLINE • done ONCE • hours on one GPU

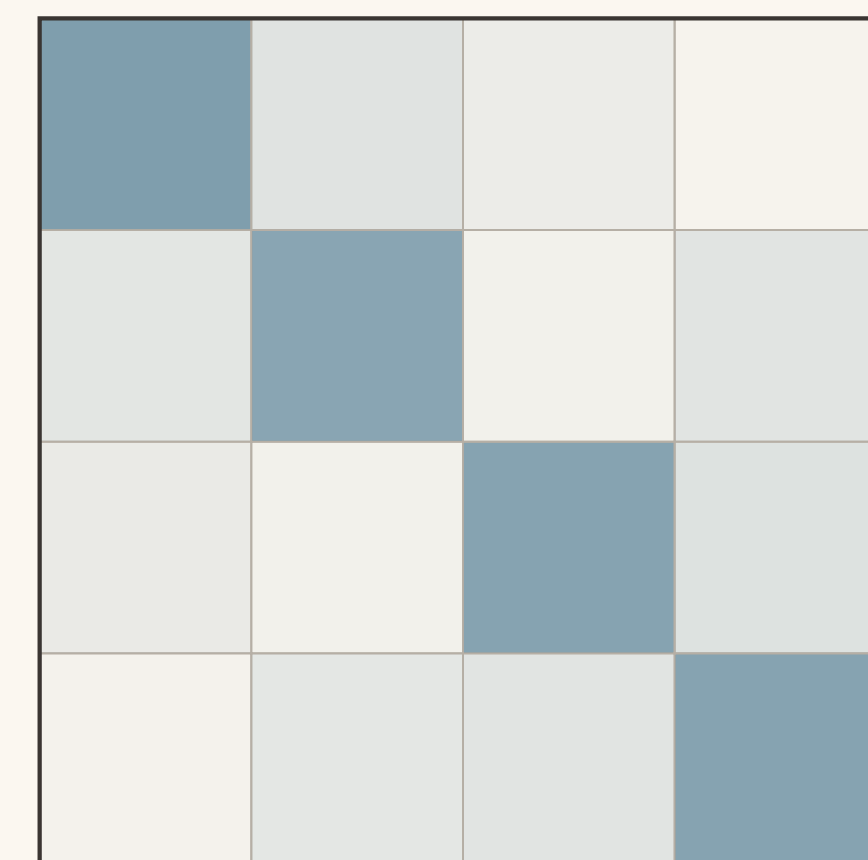


1 Sample a representation

Where the four stimuli sit in perceptual space, how noisy each is, and where the decision bounds fall — the **12 identified parameters**.

exact
forward model

stimulus

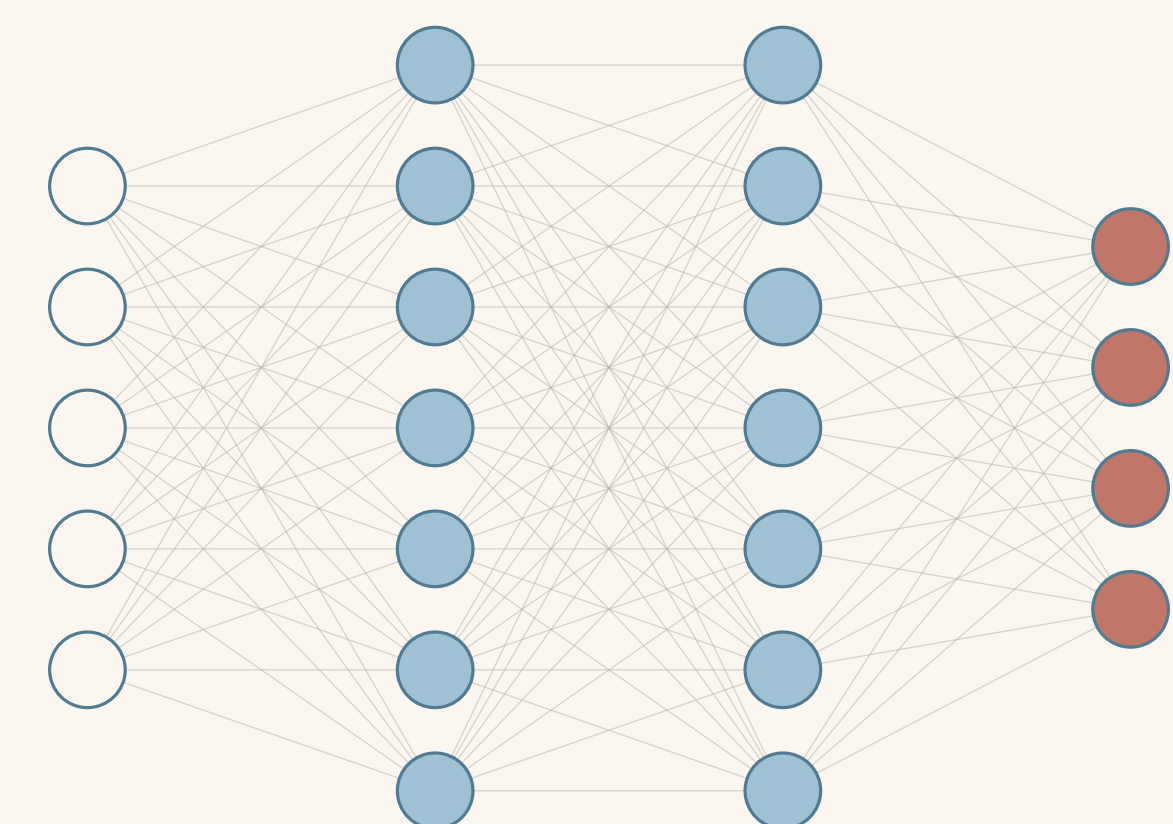


2 Simulate the data

Push θ through the exact forward model to get the matrix it would produce. Cheap — so we can make **millions** of (θ, x) pairs.

1.2×10^6
pairs

data

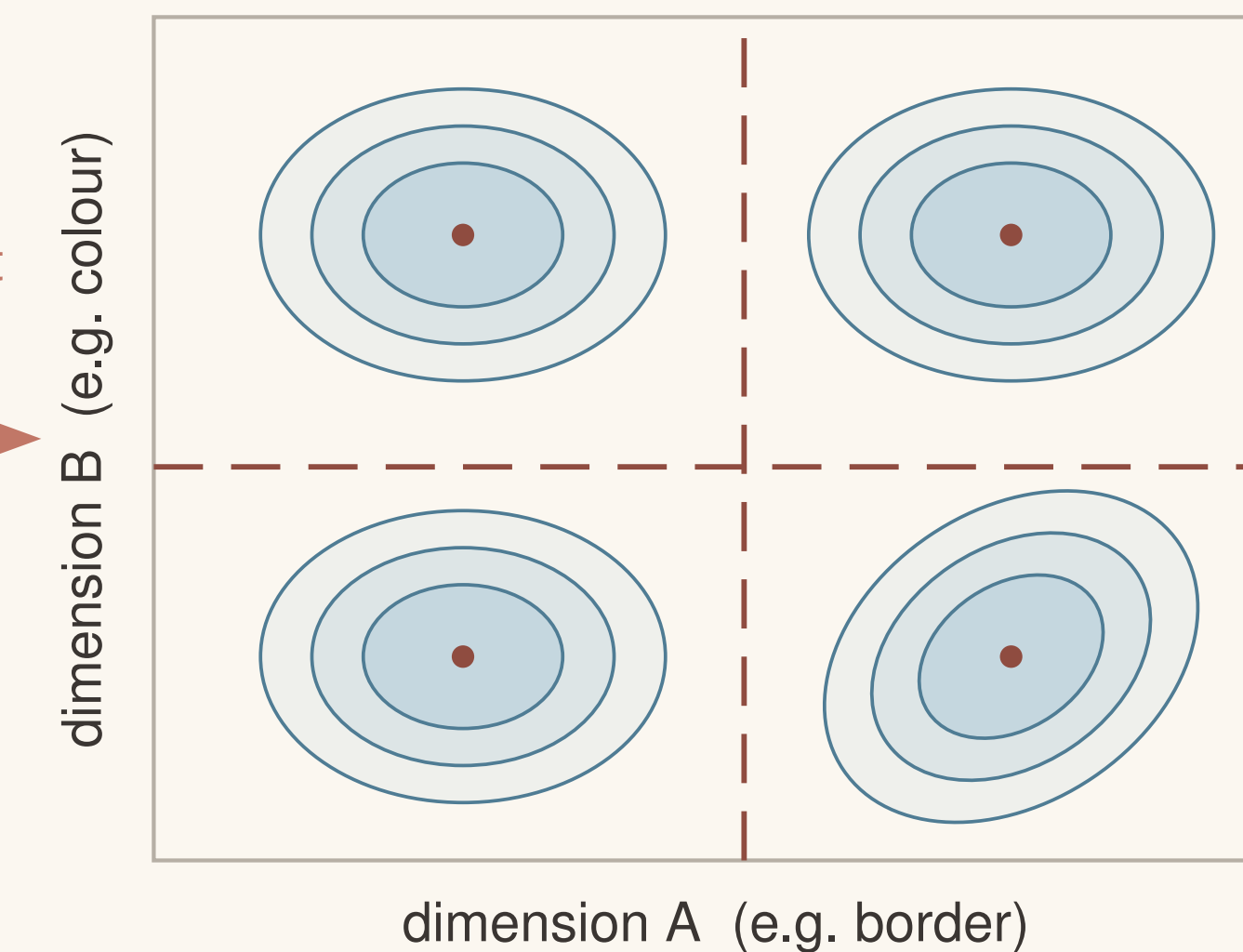


3 Learn the inverse

Train a network to map data straight back to θ , returning a **full posterior**. Expensive — but paid **once, for everyone**.

amortised: trained once,
reused for every participant

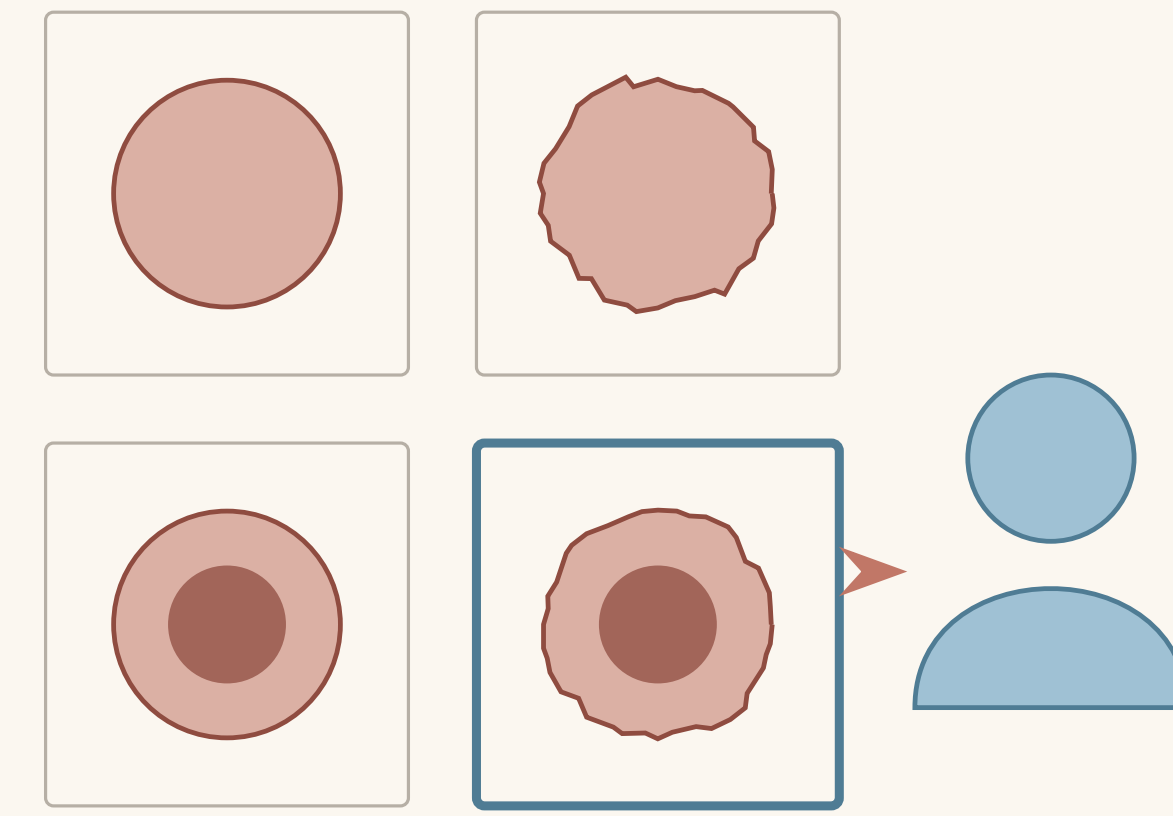
ONLINE • EVERY TRIAL • microseconds



4 Infer, in microseconds

A real matrix in, a calibrated posterior out in $\sim 3 \mu s$. MLE needs $\sim 90 ms$ and sometimes fails; this **always** answers.

where is
this person
confused?



5 Choose the next stimulus

Present the stimulus that targets the confusion we just measured — then re-infer, and go again.

every response updates the matrix, and the posterior moves with it